

Project Synopsis

Title: Enable MFA for IAM User and Federated User
Using Authenticator App

Project Domain : Cloud Computing / Amazon Web Services (AWS)

Project Type : Academic Project

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1. Abstract

This project demonstrates how to secure AWS accounts by enabling **Multi-Factor Authentication (MFA)** for IAM users and federated users. MFA requires an additional verification step beyond passwords, typically using an authenticator app. The report explains the importance of MFA, details the configuration process, and provides screenshots as evidence of implementation.

2. Introduction

Cloud computing has transformed IT infrastructure, but it also introduces risks of unauthorized access. AWS Identity and Access Management (IAM) and IAM Identity Center provide tools to manage access securely. MFA adds a second layer of protection, ensuring that even if a password is compromised, accounts remain secure.

3. Objectives

- Secure IAM and federated accounts with MFA.
- Demonstrate authenticator app integration step-by-step.
- Provide a practical guide supported by screenshots.

4. AWS IAM Overview

- **IAM Users:** Individual accounts created to access AWS resources.
- **IAM Policies:** Define permissions such as EC2 full access or S3 read-only.
- **Security Credentials:** Include console sign-in, access keys, and MFA device assignment.

5. AWS Sign-in Process

- **IAM User Login:** Requires account ID, username, and password. MFA adds a code from the authenticator app.

- **Federated User Login:** Uses Identity Center, where MFA is enforced during sign-in.

6. AWS Management Console Overview

The console home provides:

- **Dashboard:** Services, health, cost, and solutions.
- **Navigation:** Access to IAM, Identity Center, and other AWS tools.

7. IAM Identity Center

- **Dashboard:** Central hub for managing users, groups, and permissions.
- **Settings Summary:** Shows instance name, identity source, region, and organization ID.
- **Central Management:** Uses Service Control Policies (SCPs) for governance.

8. User Management in Identity Center

- **FederatedUser Creation:** Manually created with enabled status.
- **Email Verification:** Required for two-step sign-in.
- **MFA Device Assignment:** Authenticator app registered for secure login.

9. MFA Configuration

- **IAM User (MFATestUser):** MFA enabled with virtual device.
- **FederatedUser:** MFA device registered, email verification pending.
- **Authenticator App:** Generates time-based codes for login.

10. AWS Access Portal

- **Accounts Tab:** Displays linked AWS accounts (none in this case).
- **Applications Tab:** Provides access to assigned cloud applications.

11. Results and Observations

- MFA successfully enabled for IAM and federated users.
- Authenticator app provides secure login codes.
- Screenshots confirm step-by-step implementation.

12. Conclusion

Enabling MFA significantly improves AWS account security. Both IAM and federated users benefit from stronger authentication, reducing risks of unauthorized access.

13. Future Scope

- Integrate hardware MFA devices (e.g., YubiKey).
- Automate MFA enforcement across all accounts.
- Extend MFA to third-party applications via Identity Center.

15. References

- AWS Documentation: IAM and Identity Center.
- AWS Security Best Practices.
- Project screenshots (provided).